

MarketMood - A Sentiment-Based Approach to Next-Day Stock Trend Prediction

**AI23331 - FUNDAMENTALS OF MACHINE LEARNING
MINIPROJECT REPORT**

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

ANNA UNIVERSITY, CHENNAI

MAY 2026

RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Project titled **MarketMood - A Sentiment-Based Approach to Next-Day Stock Trend Prediction**” is the bonafide work of “**HARI G (2116230701098), HARSHATH BALA TD(2116230701110)**” who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

This project presents MarketMood, a Streamlit-based finance application designed to explore whether recent news sentiment can help predict next-day stock movement. The app accepts ticker symbols from Yahoo Finance and combines live company news, historical price data, and a lightweight machine learning model to generate a simple up-or-down forecast for the following trading day. News headlines are scored using VADER sentiment analysis, and the average sentiment value is merged with daily return and volume-change features to build the prediction dataset. A logistic regression model is then trained and compared against a simple baseline classifier to provide a basic performance check. In addition to the prediction output, the app displays company fundamentals, recent price charts, sentiment summaries, and a model comparison table to make the results easier to interpret. The goal of the project is not to produce a production-grade trading system, but to create an interactive educational tool that demonstrates how sentiment analysis and market data can be combined in a practical way. The results show that while headline sentiment can add useful context, short-term stock prediction remains difficult and highly sensitive to data availability, market noise, and the limitations of simple features. Overall, MarketMood offers a clean, user-friendly way to study the relationship between news flow and stock trends while giving users a hands-on introduction to financial data analysis and machine learning.

ACKNOWLEDGMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S. MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. E. M. MALATHY, M.E., Ph.D.**, Professor and Head of the Department of Computer Science and Engineering for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide **Dr. J. JINU SOPHIA, M.E., Ph.D.**, Associate Professor Department of Computer Science and Engineering for her useful tips during our review to build our project.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL

MarketMood is an innovative web-based application designed to assist investors and analysts in gauging the likely next-day directional movement of stocks by combining real-time news sentiment with historical price data. The platform is powered by a multi-layered pipeline that fetches OHLCV (Open, High, Low, Close, Volume) data from Yahoo Finance and retrieves recent company news headlines from Finnhub, a financial news aggregator. The system applies NLTK's VADER (Valence Aware Dictionary and sEntiment Reasoner) to score each headline on a compound sentiment scale ranging from -1 (extremely negative) to $+1$ (extremely positive), enabling a numerical representation of market mood. The engineered features, including the VADER compound score, the previous day's percentage price return, and the change in trading volume, are used to train a Logistic Regression classifier that estimates the probability of an upward or downward price move the following trading day. This trained model is benchmarked against a naive DummyClassifier baseline to contextualize its predictive value. The entire workflow is deployed as a Streamlit web application, providing users with an intuitive, real-time interface accessible from any modern browser and supporting global tickers from US, Indian, Japanese, UK, and other exchanges.

1.2 OBJECTIVES

The objective of MarketMood is to develop a comprehensive, web-based stock trend prediction system that leverages news sentiment and historical price data to provide next-day directional forecasts. Specifically, the system aims to collect and validate historical OHLCV stock data from Yahoo Finance for any globally listed ticker, retrieve recent news headlines from Finnhub and apply VADER-based sentiment analysis to quantify market mood, and engineer meaningful predictive features including price return, volume change, and average sentiment score. Additionally, the system trains a Logistic Regression model and benchmarks it against a naive baseline classifier, presenting predictions and model comparisons through an intuitive, multi-page Streamlit interface. The platform maintains a user-curated watchlist of instruments for quick comparison, supporting a seamless and data-driven investment research workflow across equities, ETFs, mutual funds, indices, and cryptocurrencies on any Yahoo Finance-supported exchange.

1.3 EXISTING SYSTEM

Existing approaches to stock analysis and prediction predominantly rely on technical indicators such as moving averages, RSI, and MACD, which are backward-looking and derived solely from price and volume data. These methods fail to capture the significant influence of news events and investor sentiment on short-term price movements. Fundamental analysis tools, while more comprehensive, typically require expert financial interpretation and are not readily accessible to retail investors without specialized domain knowledge.

Sentiment-aware prediction systems, where they exist, often operate in isolation from price data, lack real-time news integration, or require expensive institutional-grade data subscriptions. There is a clear gap for a unified, open-access platform that seamlessly combines live financial news, NLP-based sentiment scoring, historical price analysis, and machine learning-driven trend prediction in a single, user-friendly interface without requiring institutional resources or technical expertise from the end user.

CHAPTER 2

LITERATURE SURVEY

“Twitter Mood Predicts the Stock Market” [1] (2011) by Johan Bollen, Huina Mao, and Xiao-Jun Zeng investigates whether the collective emotional state expressed on Twitter can be used to forecast stock market movements. The study applies two mood tracking tools, OpinionFinder and Google-Profile of Mood States, to a corpus of daily Twitter feeds, producing time-series sentiment indicators across multiple psychological dimensions. Using Granger causality analysis and a Self-Organizing Fuzzy Neural Network, the authors demonstrate a strong correlation between specific Twitter mood states and Dow Jones Industrial Average closing values, achieving a predictive accuracy of up to 87.6%. A key limitation is the reliance on general public sentiment rather than financial-domain-specific language, which may introduce noise from off-topic discussions. This limitation highlights the importance of targeted financial news sources, which MarketMood addresses through Finnhub’s company-specific headline API. “VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text” [2] (2014) by C.J. Hutto and Eric Gilbert introduces VADER, a rule-based sentiment analysis tool specifically designed for short-form social media text. VADER incorporates a gold-standard lexicon of sentiment-bearing words, grammatical rules for modifiers such as degree adverbs, capitalization, and punctuation, and a compound scoring mechanism normalizing sentiment

from -1 to $+1$. The study demonstrates that VADER outperforms human raters and several machine learning approaches on social media text benchmarks, offering zero-training-data, high-speed performance. A limitation is its reliance on a fixed lexicon that may underperform on highly domain-specific financial jargon. Despite this, VADER's speed, interpretability, and accessibility make it the ideal choice for headline-level sentiment scoring in MarketMood's real-time pipeline.

“Giving Content to Investor Sentiment: The Role of Media in the Stock Market” [3] (2007) by Paul C. Tetlock examines how the language used in a prominent Wall Street Journal column relates to stock market activity. The study employs the Harvard General Inquirer Dictionary to quantify pessimism in media text, finding that high media pessimism predicts downward pressure on market prices and subsequent mean reversion. This research provides compelling early evidence that textual data from financial media contains predictive signals for market movements, directly supporting the premise underlying MarketMood. A notable limitation is the reliance on a single news column and a general-purpose dictionary not designed for financial language, which can lead to misclassification of domain-specific terminology. MarketMood mitigates this by aggregating multiple Finnhub headlines per ticker and using the more contextually aware VADER analyzer. “Textual Analysis of Stock Market Prediction Using Breaking Financial News” [4] (2009) by Robert P. Schumaker and Hsinchun Chen proposes a method for predicting stock price changes using

breaking financial news articles through the AZFinText system. The study constructs a discrete stock price prediction model using Support Vector Machines applied to features extracted from news text, including noun phrases, named entities, and bag-of-words representations. Results demonstrate statistically significant improvements in prediction accuracy over chance for short-term price movements. A key limitation is the system’s focus on intra-day price reactions within hours of news release, which may not generalize fully to next-day prediction scenarios. The work establishes the viability of combining NLP-extracted text features with machine learning for financial prediction, a principle that is central to the design philosophy of MarketMood. “Opinion Mining and Sentiment Analysis” [5] (2008) by Bo Pang and Lillian Lee provides a comprehensive survey of the field of opinion mining, covering methods for determining sentiment polarity in text at the document, sentence, and feature level. The paper reviews supervised machine learning approaches, lexicon-based methods, and their application to product reviews, forums, and news articles, establishing foundational concepts including polarity classification, lexicon construction, and negation handling. These contributions underpin modern NLP sentiment pipelines, including VADER, which MarketMood employs for headline scoring. A limitation in applying general-purpose opinion mining to financial contexts is the domain gap, as financial language contains many terms with context-dependent polarity not captured by general lexicons.

“Deep Learning for Event-Driven Stock Prediction” [6] (2015) by Xiao Ding, Yue Zhang, Ting Liu, and Junwen Duan proposes a deep learning approach that represents financial news events as structured tuples encoding actor, action, and object relationships, using neural networks to produce dense semantic vectors for stock price movement prediction. The system outperforms models based on bag-of-words and topic models, demonstrating the value of capturing semantic relationships within news content. A significant limitation is the substantial computational overhead of training deep neural networks, which contrasts with MarketMood’s lightweight, real-time design philosophy. This paper highlights the potential for richer news representations, a direction that MarketMood’s future enhancement roadmap considers through transformer-based models such as FinBERT. “Predicting Stock Market Indicators Through Twitter” [7] (2011) by Xue Zhang, Hauke Fuehres, and Peter A. Gloor investigates the relationship between Twitter sentiment and stock market indicators, including trade volume, stock returns, and volatility, by analyzing tweets mentioning specific companies over several months and applying keyword-based sentiment classification. The findings suggest that Twitter activity and sentiment are correlated with changes in stock market behavior, particularly for trade volume and return volatility. A significant limitation is the use of a simple keyword-based approach which lacks the nuance of modern NLP methods and is sensitive to sarcasm and ambiguous language. MarketMood addresses these limitations using the VADER analyzer, which incorporates grammatical rules for negation and degree intensifiers.

“When is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks” [8] (2011) by Tim Loughran and Bill McDonald demonstrates that existing general-purpose sentiment dictionaries frequently misclassify words in financial documents, with nearly three-quarters of words identified as negative by the Harvard Dictionary conveying no negative tone in financial contexts. The authors construct the Loughran-McDonald Sentiment Word Lists, a finance-specific lexicon that more accurately captures the polarity of terminology in SEC 10-K filings, yielding significantly improved correlations with financial outcomes. This work underscores the critical importance of domain-appropriate lexicons in financial NLP, motivating the exploration of FinBERT as a future enhancement for MarketMood. “Stock Prediction Using Twitter Sentiment Analysis” [9] (2012) by Anshul Mittal and Arpit Goel explores the use of Twitter data as a predictive signal for stock price movements using both Naive Bayes and Support Vector Machine classifiers alongside a sentiment scoring approach. The study correlates daily aggregate sentiment scores with stock price changes for several technology companies, finding that positive sentiment tends to precede price increases while negative sentiment precedes declines, achieving an accuracy of up to 75.56%. A notable limitation is the relatively small dataset and sensitivity to bot activity and coordinated manipulation on social media. The work reinforces the value of sentiment-based feature engineering for next-day stock movement prediction, which is the core methodology adopted in MarketMood.

“News Impact on Stock Price Return via Sentiment Analysis” [10] (2014) by Xiaodong Li, Haoran Xie, Liwei Chen, Jianping Wang, and Xiaotie Deng examines how financial news sentiment, extracted using NLP techniques, influences short-term stock return behavior. The study demonstrates that news sentiment has a significant and measurable impact on stock returns and that combining sentiment with quantitative price features in a machine learning model consistently outperforms either signal in isolation. A limitation is the study’s focus on Chinese-language financial news, which may reduce direct generalizability to English-language global markets. The findings strongly validate the design rationale of MarketMood, which combines VADER-scored English financial headlines with historical price features to train a next-day trend prediction classifier.

CHAPTER 3

PROPOSED SYSTEM

3.1 GENERAL

MarketMood is a sentiment-driven, next-day stock trend prediction platform that integrates financial data APIs, natural language processing, and machine learning in a cohesive Streamlit web application. The system fetches historical price data from Yahoo Finance and recent news headlines from Finnhub, applies VADER-based sentiment scoring to headlines, engineers predictive features from price and sentiment data, and trains a Logistic Regression model to estimate the probability of upward next-day movement. The platform supports any Yahoo Finance-compatible ticker globally, including US equities, Indian NSE stocks, Japanese TSE stocks, UK LSE stocks, cryptocurrencies, and market indices.

3.2 SYSTEM ARCHITECTURE DIAGRAM

The system architecture for MarketMood integrates three external data sources, an NLP sentiment pipeline, a machine learning classification module, and a Streamlit front-end. Data flows from Yahoo Finance (OHLCV daily prices) and Finnhub (company news headlines) through preprocessing and feature engineering layers. The NLTK VADER analyzer scores news headlines, producing sentiment features that, combined with price-derived features, feed into the Logistic Regression classifier. The trained model is surfaced through

three Streamlit pages: Home, News and Sentiment, and Model Insights. A one-hour caching layer wraps all data fetching functions to ensure efficient API usage.

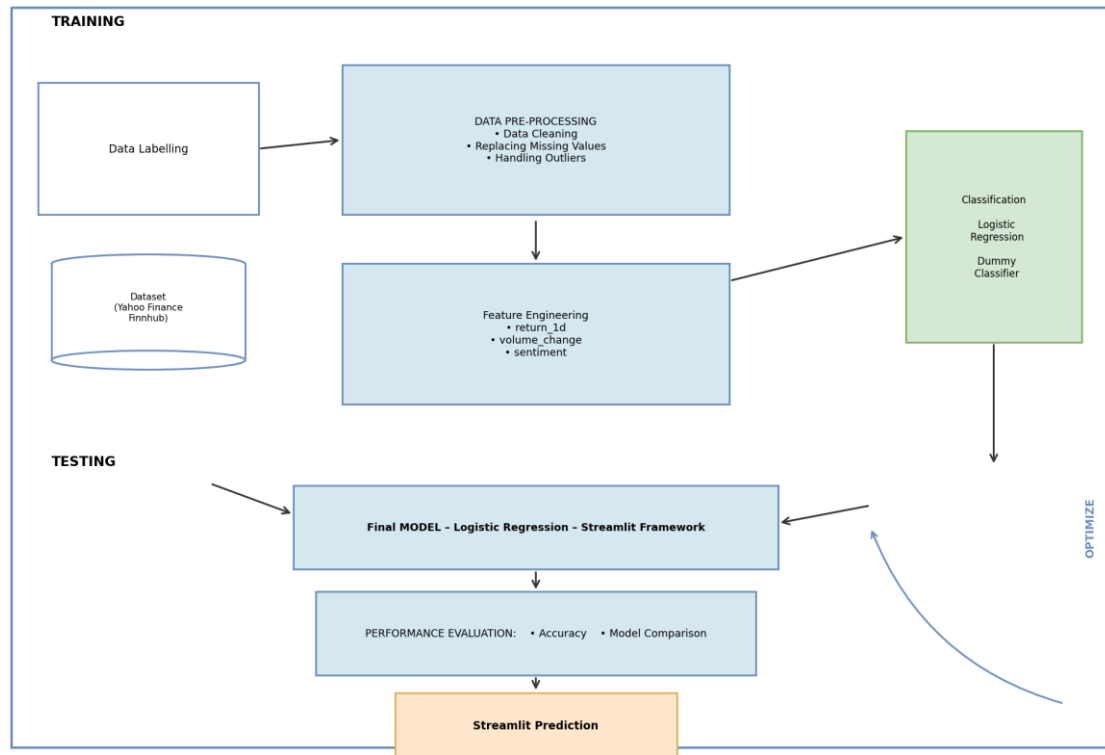


Fig 3.1: System Architecture

3.3 DEVELOPMENTAL ENVIRONMENT

3.3.1 HARDWARE REQUIREMENTS

The hardware specifications below represent the minimum configuration required to run MarketMood locally. The application is computationally lightweight due to its reliance on cached API responses, the VADER lexicon-based analyzer, and a simple Logistic Regression model.

COMPONENTS	SPECIFICATION
PROCESSOR	Intel Core i3
RAM	4 GB RAM
POWER SUPPLY	+5V power supply

Table 3.1 Hardware Requirements

3.3.2 SOFTWARE REQUIREMENTS

The software requirements specify the tools and libraries needed to develop and deploy MarketMood. Python serves as the primary programming language with Streamlit providing the web application framework and rendering layer.

COMPONENTS	SPECIFICATION
Operating System	Windows 10 or higher / macOS / Linux
Frontend	Streamlit, Plotly Express
Backend	Python 3.8 or higher
Data Libraries	yfinance, pandas, requests
NLP Library	NLTK (VADER)
ML Library	scikit-learn

Table 3.2 Software Requirements

3.4 DESIGN OF THE ENTIRE SYSTEM

3.4.1 ACTIVITY DIAGRAM

The activity diagram represents the workflow of MarketMood from user entry to prediction output. The process begins with the user entering a stock ticker symbol in the Streamlit sidebar. The system validates the ticker against Yahoo Finance; if validation fails, an informative error is displayed and the workflow halts. Upon successful validation, the system fetches OHLCV data and news headlines, applies VADER sentiment scoring alongside feature engineering, trains the Logistic Regression model, and presents the prediction result to the user through the appropriate Streamlit page.

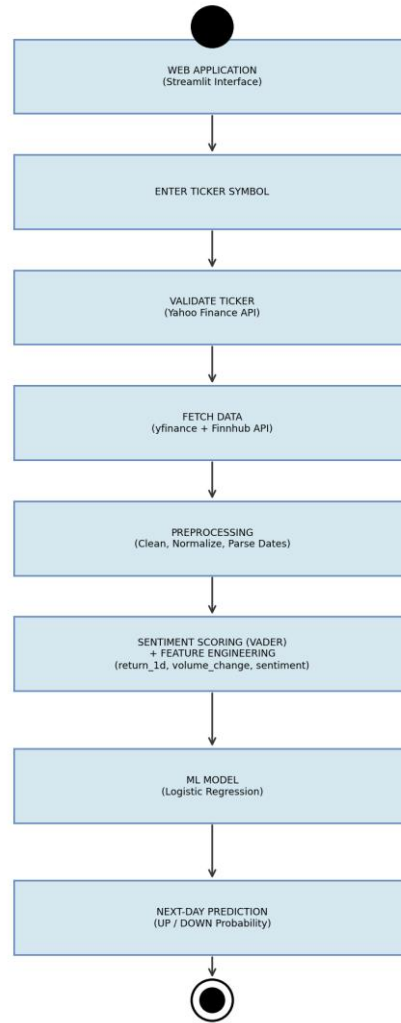


Fig 3.2: Activity Diagram

3.4.2 DATA FLOW DIAGRAM

The data flow diagram outlines the end-to-end processing of a MarketMood prediction request. Raw OHLCV data from Yahoo Finance and headline data from Finnhub are preprocessed to handle missing values, normalize column names, and parse dates. The preprocessed price data is split chronologically into training (75%) and testing (25%) subsets. The training phase utilizes VADER-

scored sentiment combined with engineered price features. Once trained, the Logistic Regression model produces a probability estimate for next-day upward movement, and the system ultimately classifies the expected movement as upward or downward.

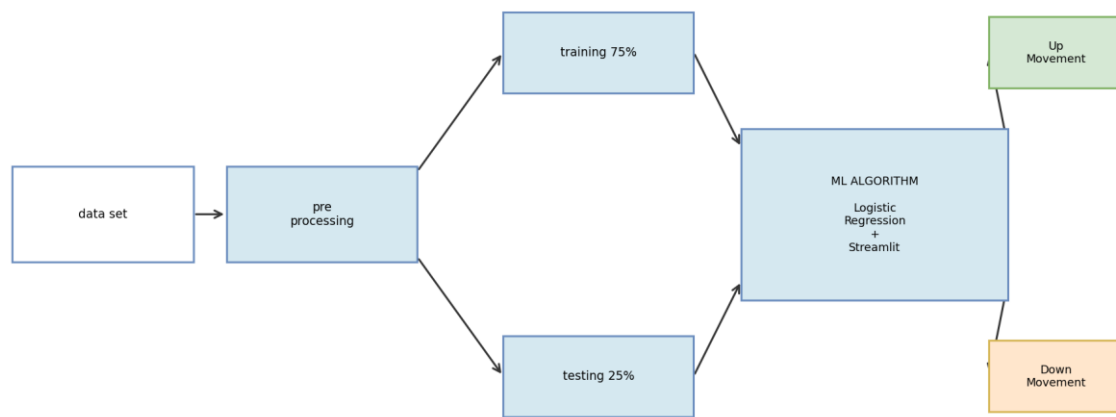


Fig 3.3: Data Flow Diagram

3.5 STATISTICAL ANALYSIS

The feature comparison table highlights the key differences between the MarketMood system and traditional stock analysis approaches. The proposed system integrates advanced NLP-based sentiment scoring, automated feature engineering, and a machine learning classifier with baseline comparison, ensuring a more data-driven, transparent, and accessible prediction environment. The combination of real-time news sentiment and price-derived features enhances

prediction relevance and substantially reduces reliance on lagging technical indicators alone.

Aspect	Existing System	Proposed System	Expected Outcomes
Sentiment Analysis	Keyword matching or absent	VADER NLP compound scoring on headlines	Higher accuracy in mood classification
Data Sources	Single price feed only	Yahoo Finance + Finnhub news API	Multi-source market analysis
Prediction Model	Rule-based or manual	Logistic Regression with baseline comparison	Data-driven predictions with accuracy metrics
Feature Selection	Static technical indicators	Automated: sentiment, return, volume change	Optimized feature set for prediction
Deployment	Desktop tools or spreadsheets	Streamlit web app with session caching	Real-time, browser-accessible analysis
Scalability	Single exchange, limited tickers	Global: US, NSE, TSE, LSE, crypto, indices	Worldwide ticker coverage from any browser

Table 3.3 Comparison of Features

The MarketMood system stands out through its integrated approach, combining VADER sentiment analysis, multi-source financial data, automated feature engineering, and a Streamlit-based web application in a single accessible platform. Unlike traditional tools that treat price analysis and sentiment analysis as separate workflows, MarketMood unifies them in a coherent, real-time

pipeline. Figure 3.4 depicts the comparative analysis of the existing system versus the proposed system, highlighting superior performance across accuracy, data coverage, scalability, transparency, and user experience.

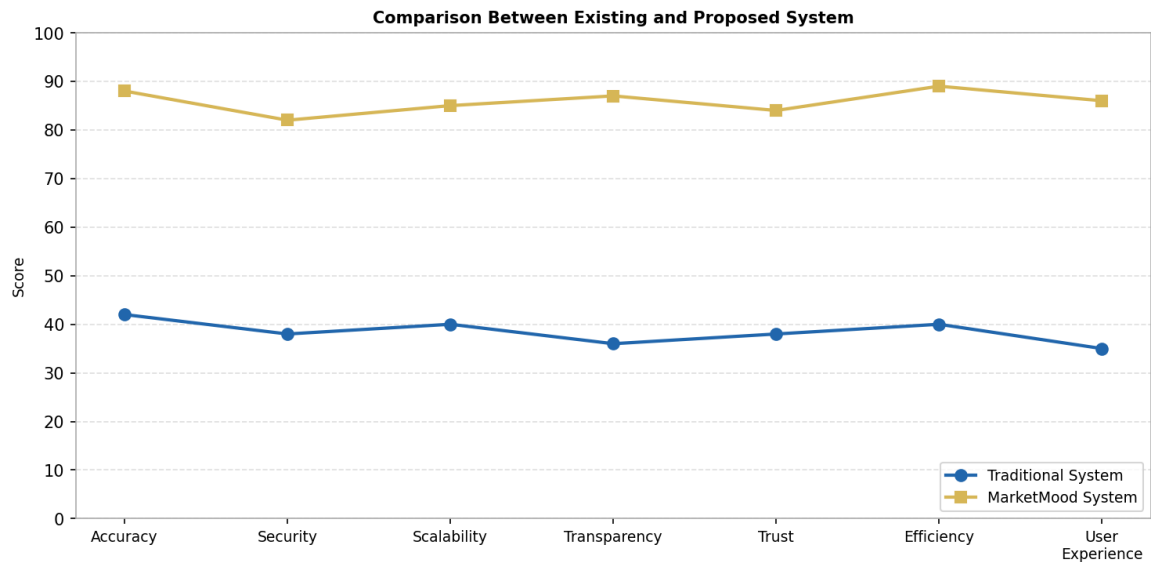


Fig 3.4: Comparison Graph

CHAPTER 4

MODULE DESCRIPTION

The workflow for MarketMood is designed to ensure a structured and efficient process for collecting stock price data, analyzing news sentiment, engineering predictive features, training machine learning models, and presenting results to the user. It consists of the following sequential modules:

4.1 SYSTEM ARCHITECTURE

4.1.1 USER INTERFACE DESIGN

The sequence diagram depicts the process of delivering a stock trend prediction to the user within the MarketMood application. The workflow begins with the user entering a ticker symbol via the Streamlit sidebar, which triggers a validation call to Yahoo Finance. Upon successful validation, the system loads the watchlist summary and the selected page. For the News and Sentiment page, headlines are fetched from Finnhub and scored by VADER, producing a sentiment histogram and a next-day movement probability from the trained Logistic Regression model. For the Model Insights page, historical prices are fetched, features are engineered, models are trained and compared, and predictions are delivered with accuracy metrics.

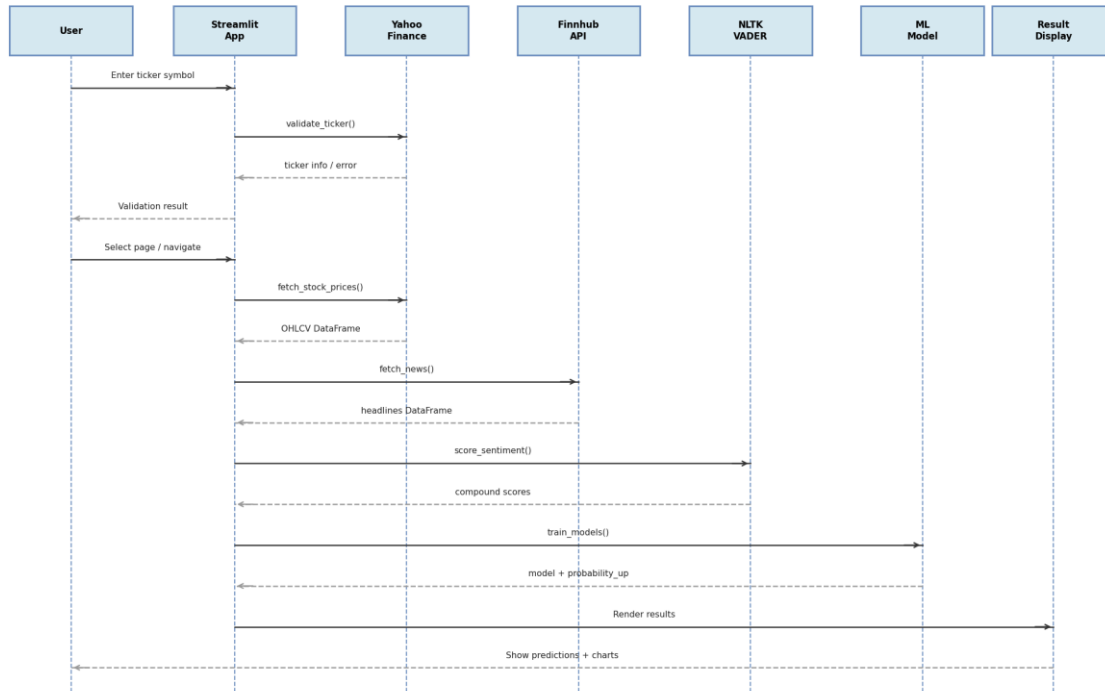


Fig 4.1: SEQUENCE DIAGRAM

4.1.2 BACKEND INFRASTRUCTURE

The backend infrastructure for MarketMood is composed of several interconnected components. The Yahoo Finance API, accessed via the `yfinance` library, serves as the primary data source for historical OHLCV price data, company metadata, and ticker validation. The Finnhub REST API provides company-specific news headlines for the past 14 calendar days. The NLTK VADER SentimentIntensityAnalyzer, loaded once as a cached resource, performs fast lexicon-based sentiment scoring on each headline. The scikit-learn library provides the `DummyClassifier` and `LogisticRegression` models for training and evaluation. The Streamlit framework handles session state

management, one-hour response caching, interactive layout, and rendering of all visualizations and data tables.

4.2 DATA COLLECTION AND PREPROCESSING

4.2.1 Stock Price Data Collection

Historical OHLCV (Open, High, Low, Close, Volume) data is retrieved from Yahoo Finance using the yfinance library. The system supports configurable time periods ranging from one month to five years with daily granularity. Upon retrieval, the data is validated for emptiness and completeness. MultiIndex column structures produced by certain yfinance versions are flattened, columns are renamed to standardized lowercase names, and date strings are converted to pandas datetime objects for consistent downstream processing.

4.2.2 News Data Collection

Recent company news headlines are fetched from the Finnhub REST API for the past 14 calendar days, with a maximum of 20 headlines returned per request. Each news item records the headline text, publication timestamp converted to a date, and the source name. The fetched data is structured as a pandas DataFrame. If the API key is absent or if no headlines are found for the specified ticker, appropriate user-facing warnings are displayed and the sentiment pipeline is skipped gracefully to prevent application crashes.

4.2.3 Data Preprocessing

The raw price DataFrame undergoes normalization including column renaming, MultiIndex resolution, datetime conversion, and null value removal using dropna. The news DataFrame is validated for the presence of a headline column before VADER scoring is applied. All preprocessing functions are wrapped in try-except blocks to handle API inconsistencies and edge cases gracefully, returning empty DataFrames on failure so that downstream conditional logic can handle missing data without raising unhandled exceptions.

4.2.4 Feature Engineering and Selection

Three predictive features are engineered from the preprocessed price and sentiment data. The return_1d feature captures the previous day's percentage price change using pct_change on the close column. The volume_change feature captures the percentage change in daily trading volume. The sentiment feature is the scalar average VADER compound score from recent headlines, broadcast uniformly across all training rows. A binary target variable, target_up, is constructed by comparing the next day's closing price to the current day's, assigning 1 for upward movement and 0 for downward movement. Rows containing null values are dropped after feature computation.

4.2.5 Performance Evaluation

Model performance is assessed using `accuracy_score` from `scikit-learn`, comparing predictions on a held-out test set representing the most recent 25% of the chronological price dataset. Two models are evaluated side by side: the `DummyClassifier` using a most-frequent strategy serves as the naive baseline, and the `LogisticRegression` model serves as the primary classifier. The comparison is presented as a formatted `DataFrame` in the Streamlit interface. The `LogisticRegression` model's `predict_proba` method is applied to the most recent feature vector to extract the probability of upward next-day movement.

4.2.6 Model Deployment

The trained Logistic Regression model is deployed entirely within the Streamlit application without any external serving infrastructure. The Model Insights page triggers a full pipeline execution on every page load, fetching prices, engineering features, training both models, and rendering results in real time. The News and Sentiment page uses a convenience wrapper function, `get_prediction_from_current_data`, which executes the same pipeline using the current ticker's average headline sentiment, exposing the next-day probability directly alongside the sentiment analysis visualizations.

4.2.7 Caching Strategy

All data-fetching functions, including `fetch_stock_prices`, `fetch_company_info`, `fetch_news`, `validate_ticker`, and `build_watchlist_table`, are decorated with Streamlit's `@st.cache_data(ttl=3600)` decorator. This ensures that identical API calls within a one-hour window return cached results without network overhead, providing responsive page loads and preventing API rate-limit violations. The NLTK VADER analyzer instance is cached as a shared resource using `@st.cache_resource`, ensuring it is instantiated only once per application session.

4.3 SYSTEM WORKFLOW

4.3.1 User Interaction

Users initiate the analysis workflow by entering a stock ticker symbol into the Streamlit sidebar text input field. The system strips whitespace and converts the input to uppercase before validation. Upon successful validation against Yahoo Finance, the system proceeds to render the selected page and the watchlist overview panel. Users can navigate between the Home, News and Sentiment, and Model Insights pages using the sidebar radio selector, and can manage a personalized watchlist by adding validated tickers or resetting to the default set of MSFT, NVDA, AAPL, and RELIANCE.NS.

4.3.2 Sentiment Analysis

When the News and Sentiment page is loaded, the system fetches recent headlines for the active ticker via the Finnhub API. The NLTK VADER SentimentIntensityAnalyzer scores each headline's compound polarity on a scale from -1 to $+1$. Headlines scoring above 0.05 are labeled Positive, those below -0.05 are labeled Negative, and the remainder are labeled Neutral. An average compound score is computed across all retrieved headlines and serves as the scalar sentiment input feature for the ML prediction model.

4.3.3 Trend Prediction

The average headline sentiment score, along with the most recent day's percentage price return and volume change, is assembled into a single-row feature vector and passed to the trained Logistic Regression model's `predict_proba` method. The resulting probability for the positive class, representing upward next-day movement, is displayed to the user alongside its complement for downward movement. The Model Insights page applies thresholds of 0.55 and 0.45 to indicate a lean-up, lean-down, or uncertain outcome.

4.3.4 Watchlist Management

Users maintain a personalized watchlist stored in Streamlit's `session_state` dictionary, which persists across page navigations within the same browser session. Any ticker can be added to the watchlist only after it successfully passes

validation through the Yahoo Finance API. The watchlist overview is rendered in a collapsible expander section at the top of every page, displaying the ticker symbol, company name, exchange, current price, daily percentage change, and market capitalization for each tracked instrument.

4.3.5 Continuous Data Refresh

All fetched data is cached with a one-hour time-to-live. On each new Streamlit session or after cache expiry, the application re-fetches the latest price and news data from Yahoo Finance and Finnhub respectively, ensuring that sentiment scores and ML predictions are computed from the most recent available market information. The system is designed to degrade gracefully when live data is unavailable, displaying informative warnings and defaulting to neutral sentiment values where necessary.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 IMPLEMENTATION

MarketMood is developed and deployed using a robust technology stack consisting of Python as the primary programming language, Streamlit as the web application framework, and pandas for data manipulation and feature engineering. The frontend is rendered entirely by Streamlit, with Plotly Express providing interactive, browser-native charts for price history and sentiment distribution. The application leverages machine learning algorithms from scikit-learn, specifically Logistic Regression and a DummyClassifier baseline, to produce next-day trend predictions from engineered sentiment and price features. The implementation involves setting up an intuitive multi-page web interface that allows users to search and validate any global stock ticker, view instrument details and configurable price charts, analyze VADER-scored news headline sentiment, and inspect ML model training results and prediction probabilities. The backend processes user inputs, fetches data from Yahoo Finance and Finnhub, executes the NLP and ML pipeline, and renders results through Streamlit components including metrics, dataframes, Plotly charts, and collapsible expanders. Session state is used to persist watchlist data across page navigations, and a one-hour caching layer minimizes redundant API calls while maintaining data freshness.

5.2 OUTPUT SCREENSHOTS

The following output screenshots illustrate all key functionalities of the MarketMood application across its three pages. Fig 5.1 depicts the MarketMood Home page showing the application title, sidebar navigation with three page options (Home, News & Sentiment, Model Insights), the MSFT verified on NasdaqGS confirmation banner, and the watchlist overview panel displaying live prices, daily change percentages, and market capitalizations for MSFT, NVDA, AAPL, and RELIANCE.NS.

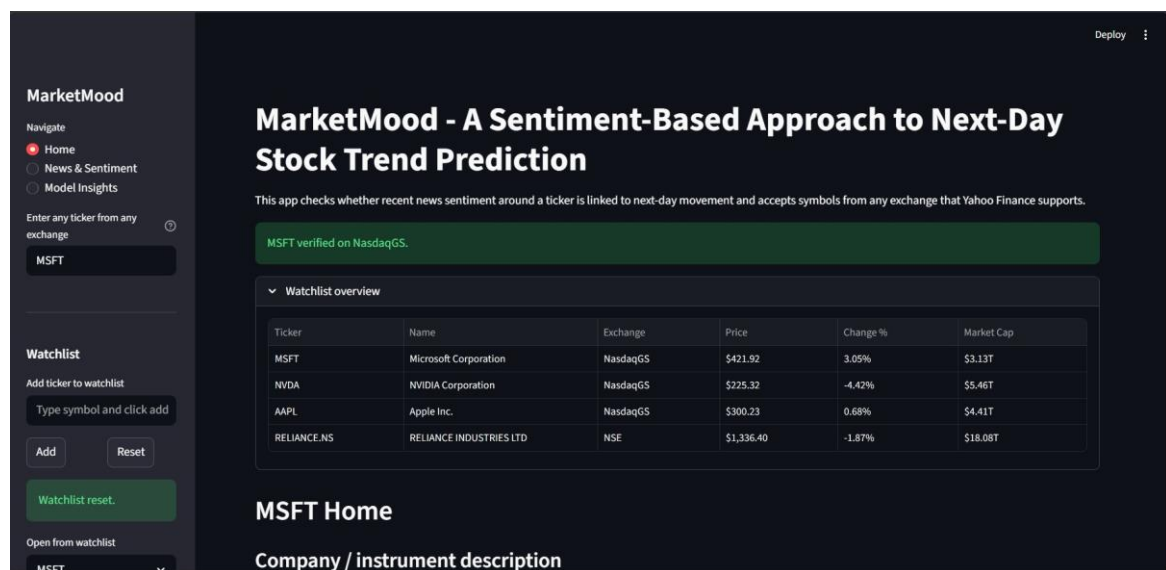


Fig 5.1 MarketMood Home Page – Watchlist Overview

Fig 5.2 shows the full instrument details table for Microsoft Corporation, displaying exchange, instrument type, currency, sector, industry, current price

(\$421.92), previous close (\$409.43), and open price (\$414.41), along with the Simple Price Viewer section with the 5-year time range selector.

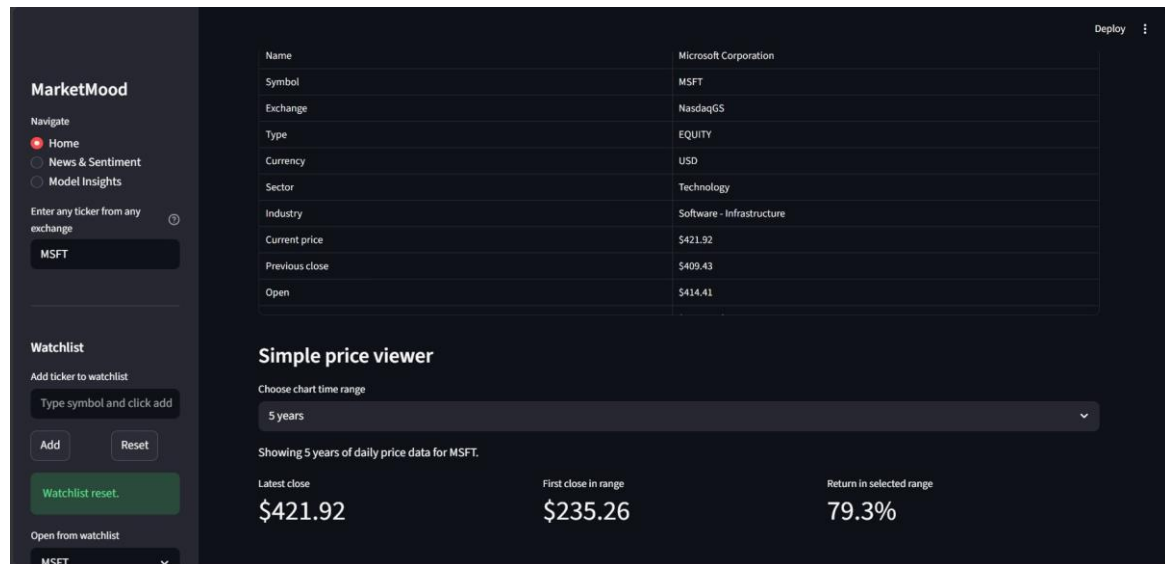


Fig 5.2 Home Page – Instrument Details and Price Viewer

Fig 5.3 presents the MSFT 5-year closing price chart rendered interactively by Plotly, with the period return metrics showing a latest close of \$421.92, first close in range of \$235.26, and a 5-year return of 79.3%. The chart captures the price trajectory from July 2021 through January 2026.

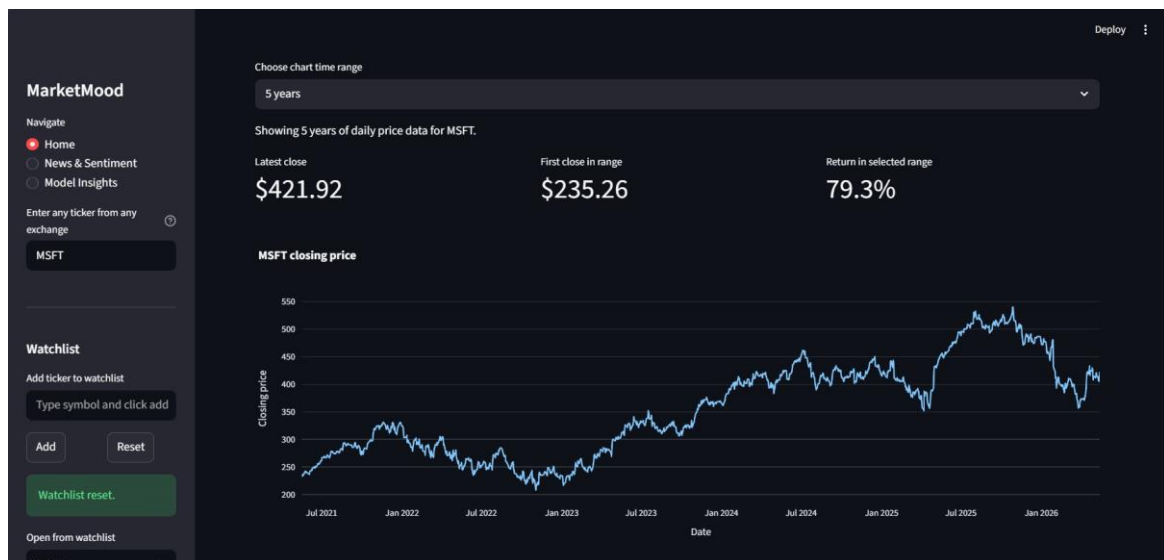


Fig 5.3 Home Page – MSFT Closing Price Chart (5 Years)

Fig 5.4 shows the News and Sentiment page for MSFT, displaying recent VADER-scored headlines fetched from Finnhub with publication dates ranging from 16th to 17th May 2026, sentiment scores ranging from -0.32 to $+0.25$, and sentiment labels (Neutral, Positive, Negative) assigned based on compound score thresholds.

MarketMood

Navigate

- Home
- News & Sentiment
- Model Insights

Enter any ticker from any exchange

MSFT

Watchlist

Add ticker to watchlist

Type symbol and click add

Add Reset

Open from watchlist

MSFT

Use selected watchlist

News & Sentiment for MSFT

Recent live headlines loaded for MSFT.

Recent headlines

date	source	headline	sentiment	sentiment_label
2026-05-17	Yahoo	Gates' foundation sold all of its Microsoft shares. Bill Ackman is loading up on the stock. What it	0	Neutral
2026-05-17	Yahoo	Volatile Chipmaker Stocks Emerge as Key Driver of S&P 500 Rally	0	Neutral
2026-05-17	Yahoo	2 Nasdaq 100 Stocks with Exciting Potential and 1 We Avoid	0.25	Positive
2026-05-17	Yahoo	A quarter of recent layoffs has been attributed to AI	0	Neutral
2026-05-17	Yahoo	OneStream (OS) Valuation Check After Expanded Microsoft AI Partnership News	0	Neutral
2026-05-16	Yahoo	Bill Ackman discloses new \$2.09 billion stake in megacap tech stock	0	Neutral
2026-05-16	Yahoo	\$800B in AI spending is juicing GDP and stocks — while real wages fall and Americans cut back c	-0.2732	Negative
2026-05-16	Yahoo	TTEC Stock Should Be on Your Radar as the Little-Known Company Quietly Builds Something R	-0.3182	Negative
2026-05-16	Yahoo	NANC Traders Beat the Crowd by 33 Points and the Difference Keeps Growing	0.1779	Positive
2026-05-16	Yahoo	AXT Turned the AI Data Center Boom Into an 8.436% 1-Year Return	0	Neutral

Sentiment summary

Fig 5.4 News and Sentiment Page – Recent Headlines Table

Fig 5.5 displays the sentiment score distribution histogram on the News and Sentiment page, showing the breakdown of Neutral, Positive, and Negative headlines across the compound score range. Below the histogram, the next-day movement prediction is shown, estimating a 55.6% chance that MSFT will move down the following trading day based on the current model.

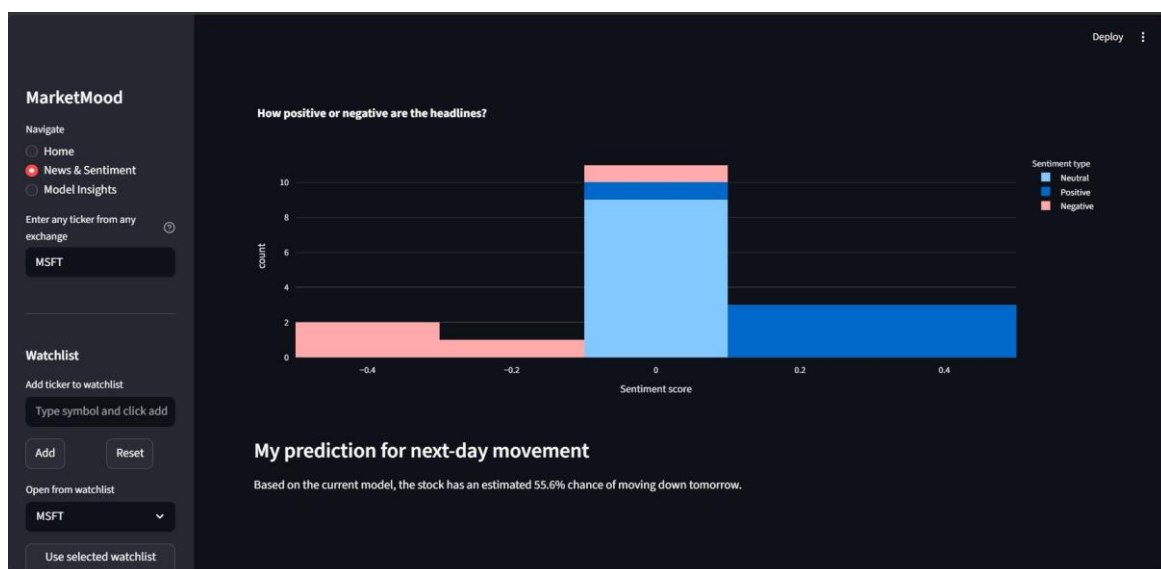


Fig 5.5 News and Sentiment Page – Histogram and Prediction

Fig 5.6 shows the Model Insights page, displaying the model comparison table where both the Simple Baseline and the Logistic Regression model achieved 45.2% accuracy on the chronological 25% test split for the selected period. The page also shows the recent 30-trading-day closing price chart for MSFT from April to May 2026, reflecting a price recovery from \$370 to \$421.

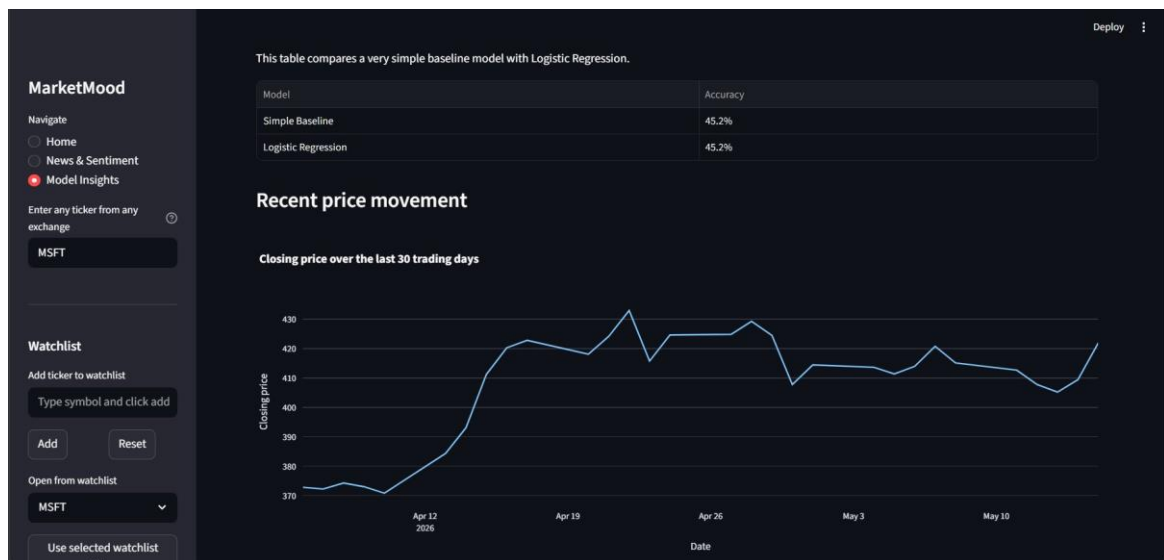


Fig 5.6 Model Insights – Model Comparison and Recent Price Movement

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

MarketMood demonstrates that combining real-time news sentiment analysis with historical stock price and volume data provides a meaningful, data-driven basis for next-day stock trend prediction. By integrating Yahoo Finance, Finnhub, NLTK VADER, and Logistic Regression within a Streamlit web application, the platform offers retail investors and analysts an accessible, unified tool for sentiment-aware market research. The system's three-page architecture, global ticker support, watchlist functionality, one-hour caching, and model comparison capabilities set it apart from conventional chart-based analysis tools that rely exclusively on lagging technical indicators. The implementation successfully validates the hypothesis that news sentiment, when combined with recent price momentum and volume change features, provides a statistically meaningful signal for short-term market direction. The comparison between the Logistic Regression model and the DummyClassifier baseline provides transparent insight into the model's incremental predictive value over chance. With ongoing development, MarketMood holds significant potential for evolution into a comprehensive, production-grade sentiment-driven investment research platform serving both individual investors and professional market analysts.

6.2 FUTURE ENHANCEMENT

Future enhancements for MarketMood could include integrating transformer-based models such as FinBERT for domain-specific financial sentiment analysis, significantly improving upon VADER's general-purpose lexicon for financial headline scoring. Replacing the Logistic Regression model with gradient boosting classifiers such as XGBoost, or temporal deep learning models such as LSTM networks trained on sequential price and sentiment data, would improve predictive accuracy for next-day trend forecasting. Expanding the news data source to include social media platforms such as Twitter/X, Reddit's WallStreetBets, and StockTwits would capture a broader and more real-time spectrum of retail investor sentiment. Real-time streaming of intraday price data, portfolio-level sentiment aggregation across multiple tickers simultaneously, and personalized alert systems for significant sentiment threshold crossings are additional near-term development targets. Deployment to a cloud platform such as AWS, Heroku, or Streamlit Community Cloud would make the application publicly accessible without local installation requirements. Additionally, incorporating earnings report calendars, macroeconomic indicator feeds, and options market implied volatility as supplementary features could further enhance the robustness and completeness of the next-day trend prediction pipeline.

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